Predicting Daily Stock Direction with End-to-End Machine Learning

A comparative machine learning study of AAPL, NFLX, MSFT, and AMZN

AAPL **NFLX** **MSFT** **AMZN**

Project Info

Course: STAT 5243 Applied Data Science
Professor: Alex Pijyan
Assignment: Project 4
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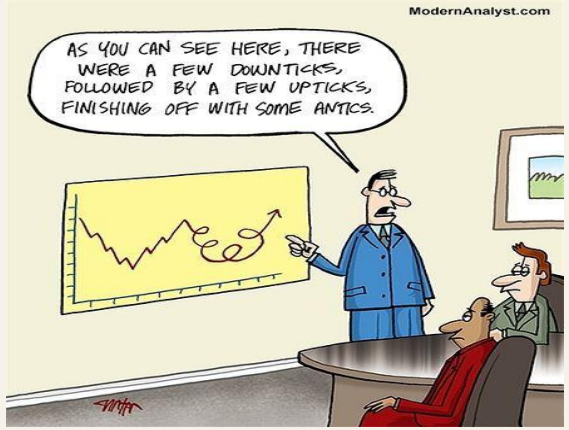
Prediction Task

Problem type: Binary Classification

Prediction unit: One stock-day

Target: UP / DOWN



In this project, we seek to predict the direction of a stock on a given day using machine learning models.

We predict direction, not exact price.

Can daily market data help predict whether tomorrow's adjusted closing price goes UP or DOWN?

AAPL | NFLX | MSFT | AMZN

Binary Target

UP or DOWN

Target rule: next-day adjusted close compared with today's adjusted close.

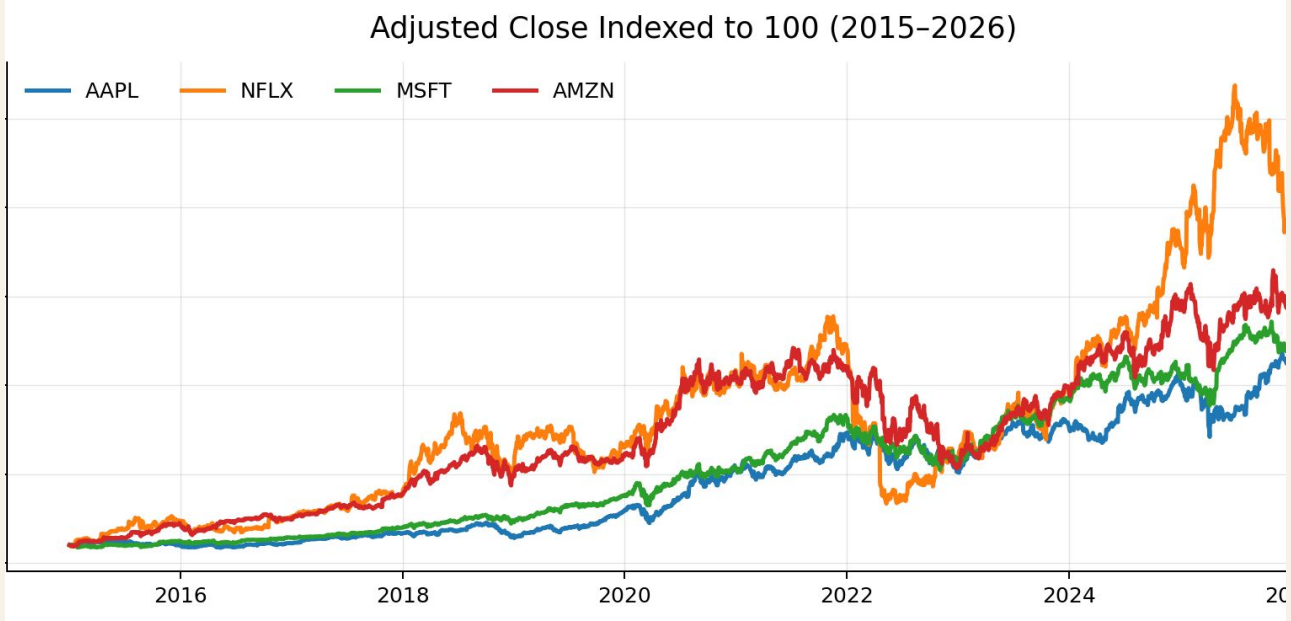
Problem Type
Classification

Prediction Unit
One stock-day

Output
UP / DOWN

We simplify stock prediction into one clear and measurable classification task.

Stock prediction is noisy, so the target must be simple.



Volatility
Daily prices can move sharply, especially during market stress.

Market noise
Short-term movements reflect many external factors not in price data.

Different behavior
The same pipeline must work across four different stocks.

Why four stocks? They are all large-cap tech names, but their price paths and volatility are visibly different.

A simple UP/DOWN target keeps the project interpretable while still testing predictive power.

We collected four matched Yahoo Finance panels.

4 stocks

AAPL, NFLX, MSFT, AMZN

2,827 rows per stock

daily observations

0 missing cells

in basic raw columns

Raw Yahoo Finance variables

Date	Trading day
Open	Opening price
High / Low	Daily price range
Close	Closing price
Adjusted	Split/dividend-adjusted close
Volume	Trading volume

Date range: January 2015 → April 2026

Basic preparation checks

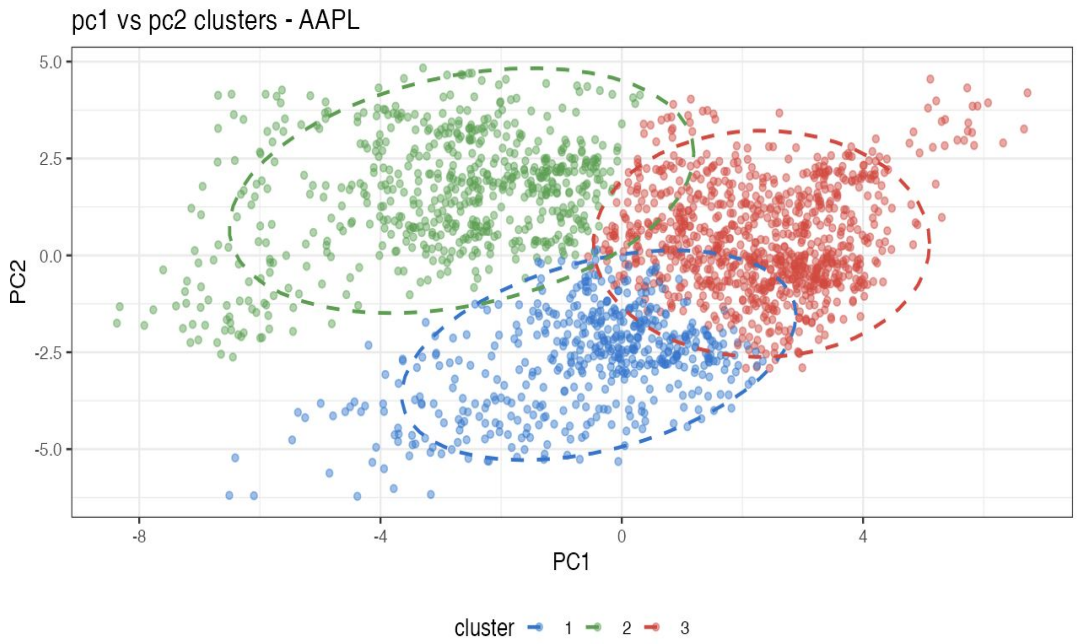
same date range across stocks • numeric price/volume columns • consistent sheet structure • next-day direction label ready for modeling

The raw data is clean and aligned, so the next step is converting prices into predictive signals.

EDA: Market Regime Structure in Technical Indicators

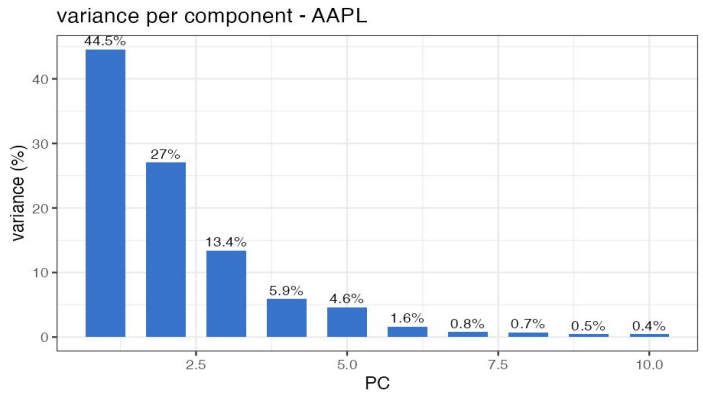
Market Data Shows Clear Grouping Patterns

PC1 VS PC2 - COLORED BY CLUSTER (K=3)



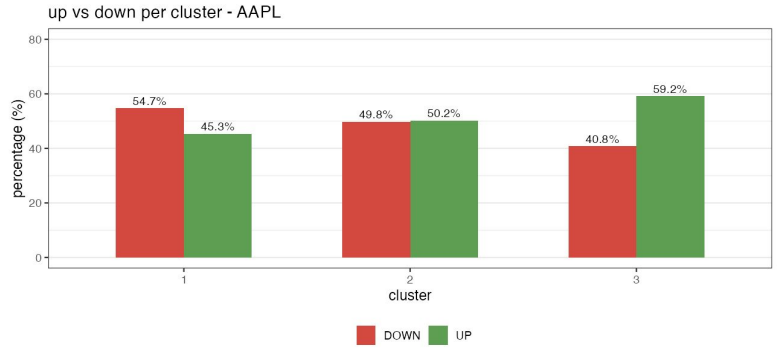
Variance Explained by Principal Components

PCA SCREE PLOT



Different Market Conditions Lead to Different Up/Down Behavior

CLUSTER VS DIRECTION (RESPONSE)



These results suggest that market conditions are not random, which motivates using clustering in our model.

Feature Engineering: PCA + K-Means Regime Labels

Analysis Pipeline

Raw Features
RSI, MACD, Ratios

PCA
Noise & Dim. Reduction

K-means Clustering
Group similar days

Market Regime
Cluster 1 / 2 / 3 Identified

Strategic Objectives & Methodology

Goal

Identify and segment distinct market regimes based on historical technical data.

Methodology Details

- Leveraged **PCA** for effective dimensionality reduction and signal clarification.
- Implemented **K-means clustering** with an optimal k=3 for regime separation.

Output Characteristics

- Daily observations are systematically assigned to a discrete cluster.
- Each cluster acts as a statistical proxy for a specific market regime (Bull, Bear, Sideways).

Why It Matters

Providing structural depth to prediction models.

- Provides critical context that raw features cannot capture in isolation.
- Enables downstream supervised models to adapt behavior based on market conditions.

PCA and k-means convert technical indicators into market regime clusters for better prediction context.

Supervised models predict daily stock direction.

Core Framework

Target: Predict UP/DOWN moves based on adjusted-close.

Inputs: Technical indicators + PCA/k-means clusters.

Validation: 80/20 chronological split with 5-fold cross-validation.

Models Compared

- **Logistic Regression:** Interpretable baseline with odds-ratio coefficients.
- **Random Forest:** Nonlinear ensemble with feature importance.
- **XGBoost:** Boosted trees for strong nonlinear prediction.

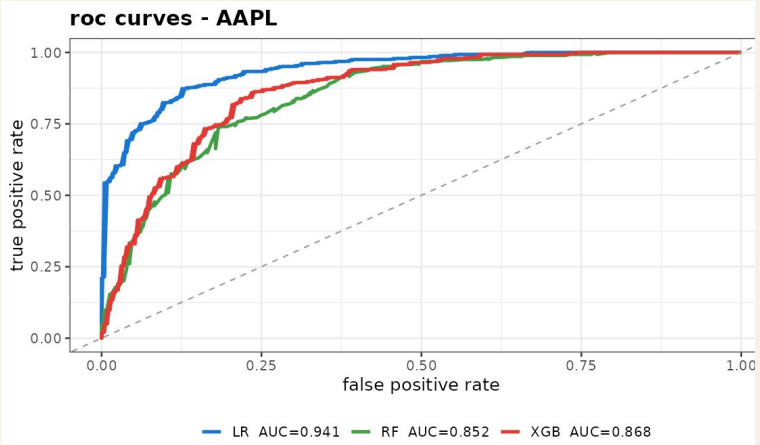
Metrics & Analysis

Evaluation Suite:

Accuracy, F1, AUC, Sensitivity, Specificity, Precision, Recall.

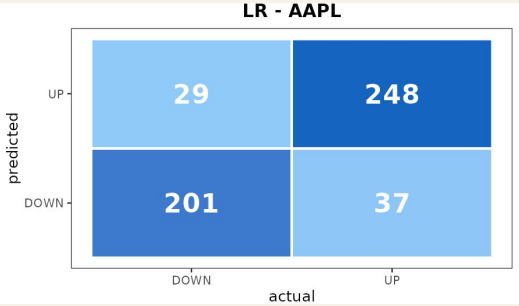
Interpretability Method:

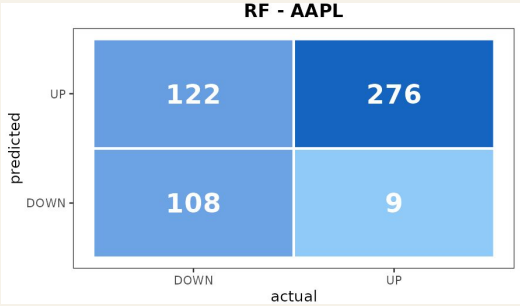
Used RF/XGBoost feature importance and Logistic Regression odds ratios to understand signal drivers.

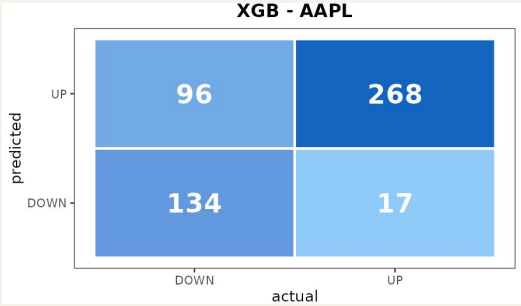


We compared an interpretable linear model (logistic regression) with two nonlinear ensemble models (random forest, XGBoost) to evaluate both predictive performance and model explainability.

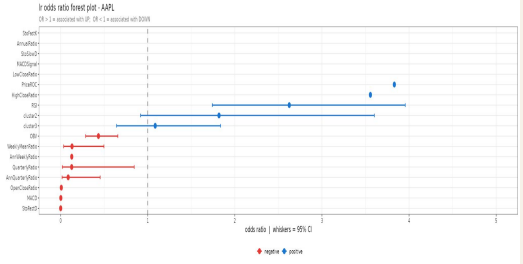
Model Confusion Matrices (Logistic Regression | Random Forest | XGBoost)

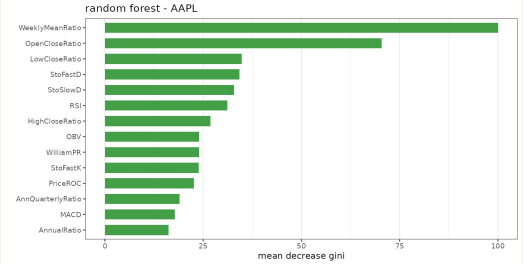


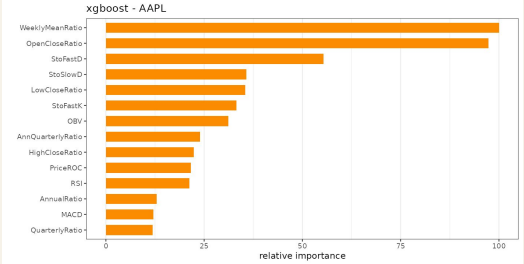




Feature Importance / Model Coefficients Analysis







Model selection is based on analysis of multiple metrics.

Final Model Selected

Logistic Regression achieved the best overall performance and stability across all four tickers: **AAPL, AMZN, MSFT, and NFLX.**

Average Scores (Top Performance)

Accuracy: 0.865

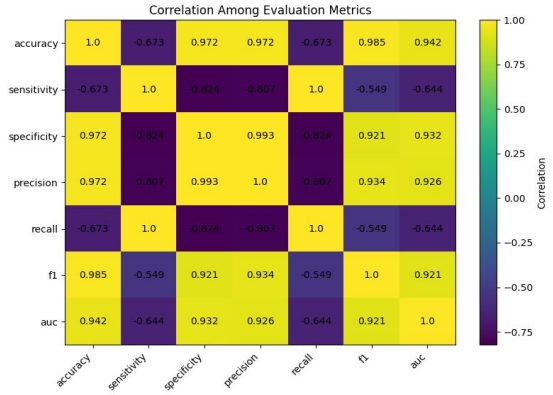
F1 Score: 0.872

AUC: 0.944

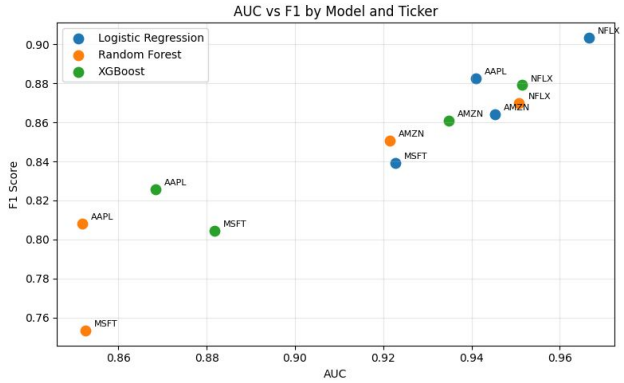
+0.335 Average Accuracy Gain over Baseline

Visual Performance Evidence

Metric-Correlation Heatmap



AUC vs. F1 Plot (Model Ranking)



Key Comparative Insights

- **Stability:** Lowest AUC variability and smallest sensitivity-specificity imbalance.
- **Tree-based Models:** Higher sensitivity but weaker specificity (often overpredicted UP days).

Chart Analysis Findings

- **Alignment:** Heatmap shows AUC, F1, accuracy, and precision are strongly aligned, ensuring rankings are robust across metrics.
- **Consistency:** AUC vs. F1 plot confirms Logistic Regression points sit in the upper-right for all tickers, indicating high performance and balance.
- **Observation:** Tree-based methods perform well for specific tickers (NFLX/AMZN) but are less consistent overall.

Logistic Regression is the best model due to its consistent ranking performance and classification balance across all examined tickers.

Using survival analysis to estimate time until a large downside move.

Methodology & Framing

- Reframe the stock-direction task as a **time-to-event** problem: we estimate how long until a large downside move (daily return of **-3%** or lower).
- Daily stock data was converted into **survival spells**, measuring trading days until the next event. **Right-censoring** was applied to spells without an observed event.

Cox Model Performance

Concordance Index: **0.65**

Top Predictors:

- 21-day rolling volatility (Hazard Ratio 1.21)
- WeeklyMeanRatio (Hazard Ratio 1.15)

Both factors significantly increase the hazard of a downside event.

Risk Stratification

Median time to event by risk group:

Low Risk: 16 days
Med Risk: 8 days
High Risk: 4 days

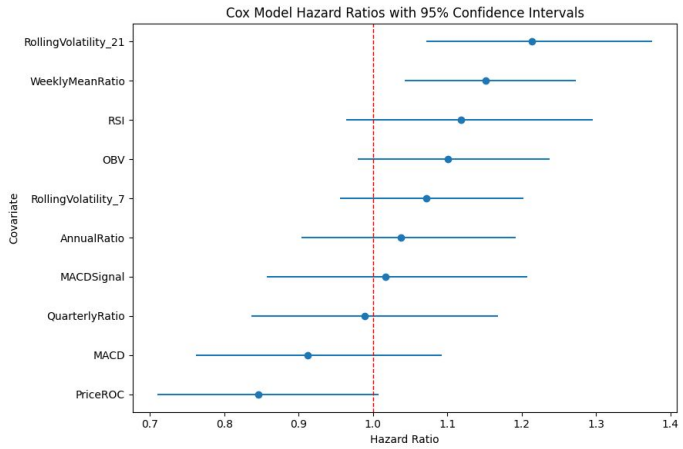
Log-rank test: p < 0.005 (highly significant)

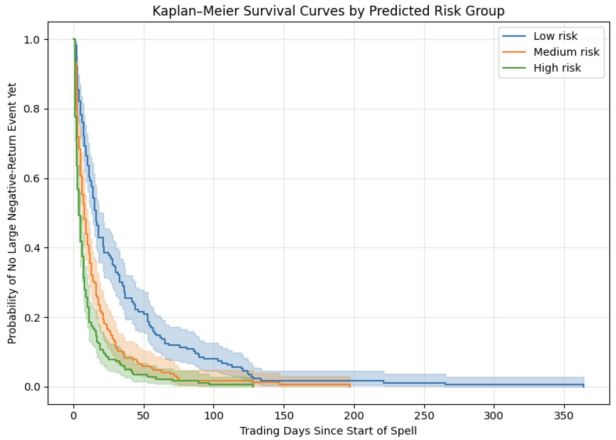
Cox proportional hazards model equation:

$$h(t \mid X) = h_0(t)\exp(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p)$$

Kaplan-Meier estimator:

$$\hat{S}(t) = \prod_{t_j \le t}\left(1 - \frac{d_j}{n_j}\right)$$





Survival analysis shows that volatility and short-term weakness signal faster arrival of large downside moves.

Final Conclusions & Future Outlook

Market Complexity

Daily stock direction is difficult to predict because short-term price movements are noisy, non-stationary, and influenced by unpredictable external events.

Modeling Scope

Real-world trading constraints were not modeled, including transaction costs, bid-ask spreads, slippage, taxes, and portfolio risk.

Generalizability

Limited scope as analysis only used four large-cap technology stocks: AAPL, NFLX, MSFT, and AMZN.

Future Roadmap

Integration of walk-forward validation, macroeconomic variables, market volatility indicators, news sentiment, and probability calibration.

ML models capture weak predictive signals, but robust financial modeling requires careful validation and realistic assumptions.